Retinal Blood Vessel Segmentation: Replicating and Improving CAR-UNet on FIVES

22AIE437 Medical Image Processing – Final Project

Group **[Group No. / Name]**

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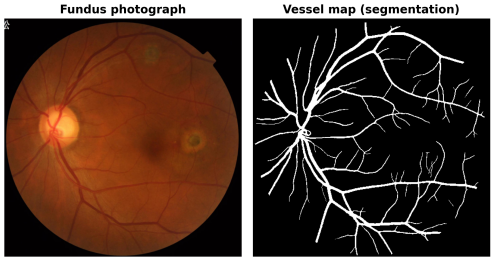
[Member 3 name] (**[Roll No.]**) **[Member 4 name]** (**[Roll No.]**)

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Problem and motivation

- The retina is the only place where doctors can look at blood vessels **without any procedure** – just a fundus photograph.
- Vessel width, branching and breaks change with **diabetic retinopathy, glaucoma, AMD and hypertension**.
- Tracing every vessel by hand is slow, tiring and differs from expert to expert.
- The **thin capillaries** change first – and are the hardest to see.
- **Why AI:** a segmentation network gives a consistent, pixel-level vessel map in about a second, so screening can scale.



Input: colour fundus photo. Output: binary vessel map (white = vessel).

Dataset and objectives

Table 1: The FIVES dataset [1] as used in this project.

Images	800 colour fundus photos, 2048×2048
Labels	pixel-wise vessel masks (expert consensus)
Groups	AMD, DR, Glaucoma, Normal (200 each)
Train	480 images (120 per group)
Validation	120 images (30 per group, stratified)
Test	200 images (50 per group) – used once
Working size	512×512 (baseline), 1024×1024 (ours)

Task: binary semantic segmentation (vessel vs. background).

Objectives

1. Replicate CAR-UNet [2] on FIVES with a fair protocol.
2. Modify it to recover thin vessels and keep vessels connected.
3. Prove every gain with an ablation and significance tests.

Metrics: Dice, IoU (main); sensitivity, specificity, accuracy, AUC (as in the paper); clDice (connectivity). Computed inside the retina (FOV), per image, over 3 seeds.

Base paper: CAR-UNet

Channel Attention Residual U-Net for Retinal Vessel Segmentation

C. Guo, M. Szemenyei, Y. Hu, W. Wang, W. Zhou, Y. Yi — **IEEE ICASSP 2021** (conference) — arXiv:2004.03702 [2]

What the paper proposes

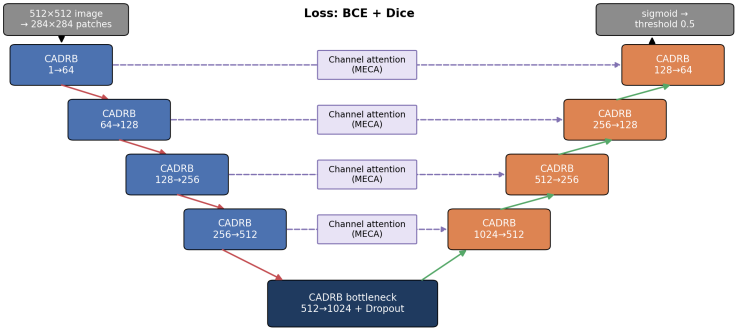
- Start from U-Net [3].
- Replace every conv block with a **residual block + channel attention** (CADRB).
- Put channel attention (MECA, based on ECA-Net [4]) on every **skip connection**.
- Trained with BCE loss and Adam.

Table 2: Results reported in the paper.

Dataset	Acc	Sen	Spe	AUC
DRIVE	0.9699	0.8135	0.9849	0.9852
CHASE_DB1	0.9751	0.8439	0.9839	0.9898
STARE	0.9743	0.8445	0.9850	0.9911

The paper does not report Dice or IoU.

Base architecture and key blocks



Inside each CADRB: conv-BN-ReLU → conv-BN → MECA → + shortcut → ReLU

284×284 input patch → 100×100 vessel map (valid convolutions, as in U-Net).

Encoder / decoder: encoder finds *what* is in the patch, decoder recovers *where*; skip connections bring back fine detail.

CADRB: conv–BN–ReLU → conv–BN → attention, plus a shortcut. The block only learns a correction, so deep training is stable.

MECA (channel attention): one weight per feature channel – strong detectors are boosted, weak ones damped.

Skip attention: encoder features are re-weighted before they reach the decoder.

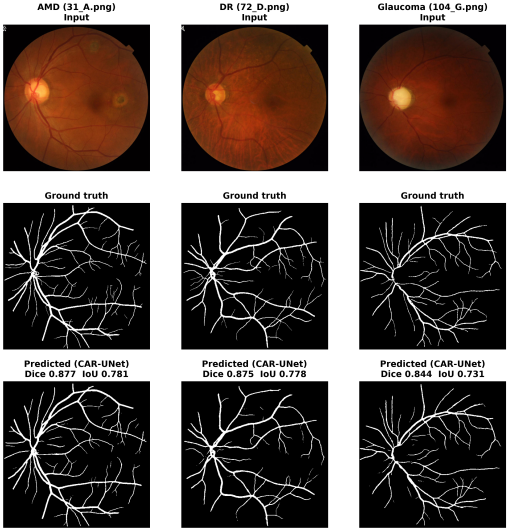
Replicated vs. reported results

Table 3: Paper results (as reported) next to our replication on FIVES. Ours: 200 test images, mean ± std over 3 seeds.

Model	Dataset	**Dice**	**IoU**	Acc	Sen	Spe	AUC
CAR-UNet – paper (reported)	DRIVE	n/r	n/r	0.9699	0.8135	0.9849	0.9852
CAR-UNet – paper (reported)	CHASE_DB1	n/r	n/r	0.9751	0.8439	0.9839	0.9898
CAR-UNet – our replication	**FIVES**	**0.8382 ± 0.0031**	**0.7295 ± 0.0043**	0.9728	0.8310	0.9855	0.9860
U-Net – our reference	FIVES	0.8344 ± 0.0034	0.7246 ± 0.0050	0.9723	0.8247	0.9852	0.9847

- Our accuracy, specificity and AUC land in the same range as the paper; sensitivity is in between its DRIVE and CHASE values.
- Numbers are not one-to-one: the paper used small single-source sets (20–40 images), we use the 800 multi-disease FIVES images required for this project.
- CAR-UNet beats U-Net on **154 / 200** test images (paired Wilcoxon, $p = 2 \times 10^{-14}$) – the replication reproduces the paper's claim.
- Honest difference: we kept U-Net's valid-convolution geometry, 64 base filters and Dropout (paper: 16 filters, DropBlock).

Replicated model: sample predictions



- Three test images from three disease groups (median-Dice cases, not hand-picked).
- Rows: **input**, **expert ground truth**, **our predicted mask**.
- Thick vessels are traced almost perfectly.
- What goes wrong: **thin capillaries** at vessel ends are missed or broken into pieces – this is what our modification targets.

Code structure and training settings

Code (PyTorch) – GitHub repository

```
src/
 preprocessing_fives.py   green, CLAHE, bilateral
 data_loader_fives.py     split, patches, augment
 car_unet.py              CAR-UNet (+ dual attn)
 losses.py                BCE + Dice (+ clDice)
 metrics.py               FOV metrics, clDice
 utils.py                 overlap-tile inference
scripts/
 train_fives.py           training loop + test
 evaluate_post.py         threshold / TTA
 summarize_*.py           tables, Wilcoxon tests
```

Pipeline: load + preprocess → random 284×284 patches → model → loss → full-image prediction (overlap-tile) → metrics.

Table 4: Training settings (identical for every model).

Framework	PyTorch 2.6, CUDA 12.4
Optimizer	Adam, weight decay 10^{-5}
Learning rate	10^{-3}, cosine decay
Batch size	8 patches (284×284)
Patches	6 fresh per image per epoch
Epochs	30 (best val-Dice checkpoint)
Loss	0.5 BCE + 0.5 Dice
Augmentation	flips, 90° rotations, elastic
Seeds	3 per model
Hardware	RTX 4050 laptop GPU (6 GB)
Time	≈91 min per run (CAR-UNet)

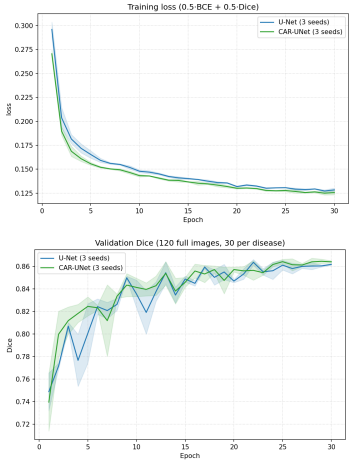
Training logs and curves

```
$ python scripts/train_fives.py --model car_unet --seed 0

CAR-UNet on FIVES | run: car_unet_seed0 | device: cuda
Parameters: 32,435,132
Train images: 480 (2880 patches/epoch) | Val images: 120 | Test images: 200
Epoch [01/30] 198.7s | lr 1.00e-03 | train loss 0.2760 | val Dice 0.7186 Sens 0.6390 Spec 0.9869 <- best
Epoch [02/30] 198.7s | lr 9.97e-04 | train loss 0.1954 | val Dice 0.7725 Sens 0.7027 Spec 0.9883 <- best
Epoch [03/30] 194.1s | lr 9.89e-04 | train loss 0.1759 | val Dice 0.8017 Sens 0.7583 Spec 0.9859 <- best
   ...   (epochs 4-26)
Epoch [27/30] 192.7s | lr 4.32e-05 | train loss 0.1255 | val Dice 0.8604 Sens 0.8665 Spec 0.9844
Epoch [28/30] 191.9s | lr 2.45e-05 | train loss 0.1253 | val Dice 0.8681 Sens 0.8763 Spec 0.9849 <- best
Epoch [29/30] 191.6s | lr 1.09e-05 | train loss 0.1247 | val Dice 0.8663 Sens 0.8710 Spec 0.9852
Epoch [30/30] 192.6s | lr 2.74e-06 | train loss 0.1245 | val Dice 0.8647 Sens 0.8659 Spec 0.9855
Training finished in 96.3 min. Best val Dice 0.8681 at epoch 28.
Loaded best checkpoint: epoch 28, val Dice 0.8681
--- test set (200 images) ---
Sensitivity: 0.8411 +/- 0.1084
F1_Dice: 0.8411 +/- 0.0899
IoU: 0.7331 +/- 0.0969
AUC_ROC: 0.9871 +/- 0.0169
```

Real training log of the replicated CAR-UNet (seed 0).

- Loss falls smoothly; no sign of overfitting.
- Validation Dice levels off near 0.86 by epoch $\sim$25, so 30 epochs is enough.
- The best epoch is picked on validation images, never on test images.



Loss and validation Dice, mean $\pm$ std over 3 seeds (U-Net vs. CAR-UNet).

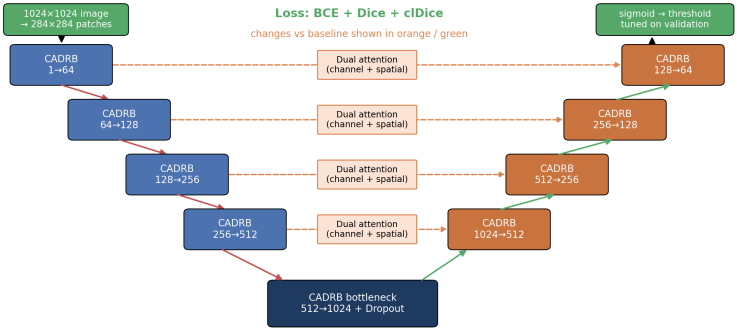
What we changed – and why

Table 5: Each modification targets one weakness of the base paper.

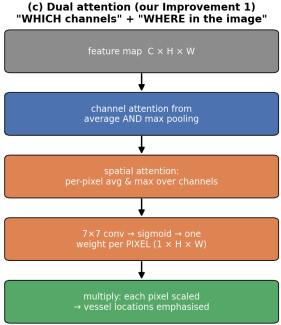
Weakness of CAR-UNet	**Our modification**	**Type**
Channel attention says *which* features matter, but not *where* the vessels are	**1. Dual attention:** channel attention (avg + max pooling) + spatial attention (CBAM [5]) in all 13 attention modules	new attention module
BCE counts pixels, so a broken thin vessel costs almost nothing	**2. clDice loss** [6]: compares vessel skeletons, rewards connected vessels	loss function
At 512×512 the finest capillaries are erased	**3. 1024×1024 working resolution**, scored on the same 512 ground truth	input resolution
Fixed 0.5 decision threshold	**4. Threshold chosen on validation** (flip TTA tested too)	prediction

Everything else – data split, seeds, optimizer, epochs, augmentation – is kept identical to the baseline.

Modified architecture



Inside each CADRB: conv-BN-ReLU → conv-BN → Dual attention → + shortcut → ReLU

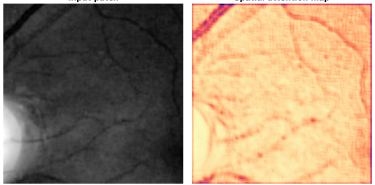


Changes vs. the baseline in orange/green: dual attention everywhere, 1024 input, clDice loss, tuned threshold.

Dual attention: per-channel weights *and* one weight per pixel. Only **+1,274 parameters**.

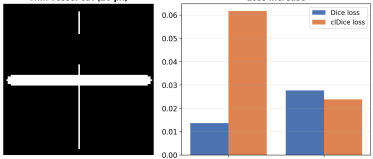
How each modification works

Dual attention



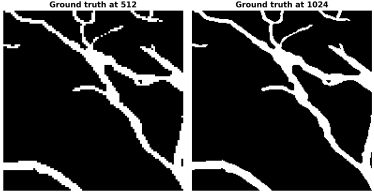
The spatial attention map traces the vessel tree, thin vessels included – the network now knows *where* vessels are.

clDice loss



Cutting a thin vessel (20 px) raises the clDice loss **4.5**× more than the Dice loss; trimming a thick vessel costs less. It punishes exactly the error we care about.

1024 resolution



At 512 thin vessels in the ground truth are already broken into pieces; at 1024 they are continuous. The model can only learn what it can see.

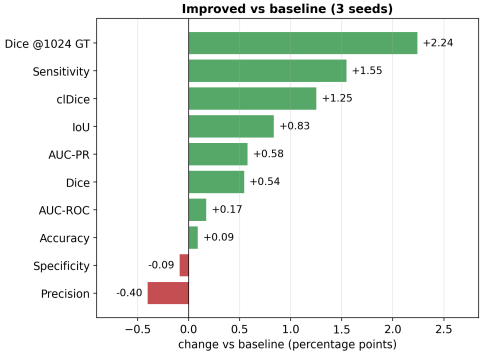
Threshold: the 1024 models are slightly cautious, so validation picks 0.44–0.48 instead of 0.5, recovering faint vessels.

Comparison: baseline vs. modified model

Table 6: Same data split, seeds and settings. 200 test images, mean over 3 seeds; p = paired Wilcoxon test.

Metric	Baseline	Modified	Change	p
Dice	0.8382 ± .0031	**0.8436 ± .0027**	**+0.54**	5×10^{-10}
IoU	0.7295 ± .0043	**0.7379 ± .0039**	**+0.83**	2×10^{-10}
Sensitivity	0.8310	**0.8465**	**+1.55**	7×10^{-19}
Specificity	0.9855	0.9846	−0.09	6×10^{-15}
Accuracy	0.9728	**0.9737**	**+0.09**	2×10^{-8}
AUC-ROC	0.9860	**0.9878**	**+0.17**	1×10^{-21}
clDice	0.8626 ± .0057	**0.8751 ± .0005**	**+1.25**	5×10^{-25}

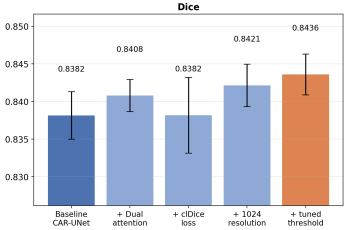
Change in percentage points. Dice better on 140/200 images, connectivity on 175/200; better in all 3 seeds.

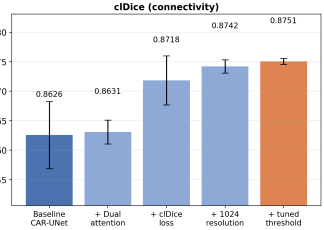


Ablation: what each component contributes

Table 7: Adding one modification at a time (3 seeds each, mean ± std).

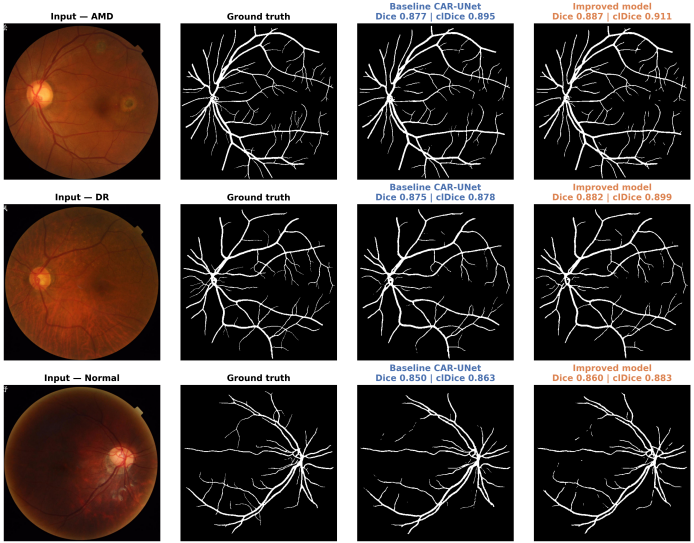
Step	Dice	IoU	Sensitivity	Precision	clDice
Baseline CAR-UNet	0.8382 ± .0031	0.7295	0.8310	0.8557	0.8626
+ Dual attention	0.8408 ± .0021	0.7339	0.8308	**0.8620**	0.8631
+ clDice loss	0.8382 ± .0050	0.7293	0.8432	0.8430	0.8718
+ 1024 resolution	0.8421 ± .0028	0.7359	0.8326	**0.8632**	0.8742
+ Tuned threshold (final)	**0.8436 ± .0027**	**0.7379**	**0.8465**	0.8516	**0.8751**





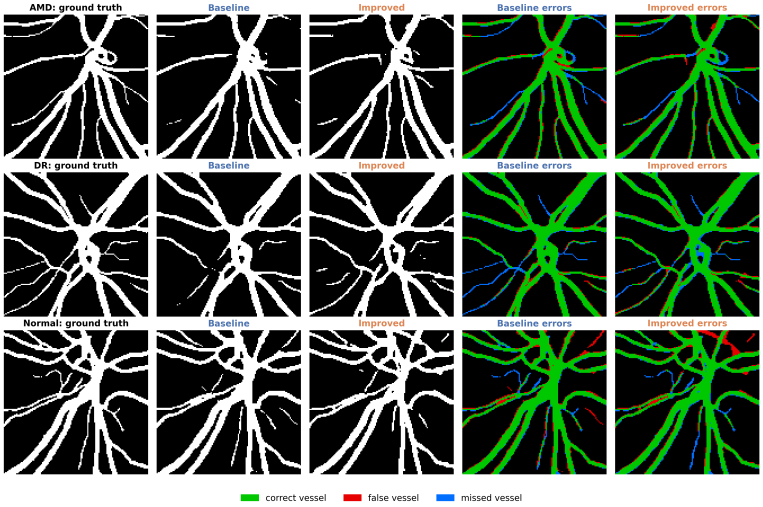
Dual attention: fewer false vessels, most stable.
clDice: big jump in connectivity and sensitivity; Dice flat (more detections at 0.5).
1024: Dice and precision up, thin vessels kept.
Threshold: turns that precision into more vessels found.

Qualitative results: side by side



- Same test images, same ground truth, both models run live.
- Picked as the median case of each disease – not cherry-picked.
- The improved model scores higher on every example (Dice and clDice).
- Differences are small at this scale – they show up on the thin vessels (next slide).

Qualitative results: thin vessels up close



- Zoom on the 128×128 region with the most thin vessels (picked from the ground truth).
- **Blue** = missed vessel, **red** = false vessel, **green** = correct.
- The improved model leaves **fewer blue gaps**: thin branches are found and stay connected.
- A few extra red pixels – the precision cost we see in the table.

Analysis: where the gains come from – and what did not work

Table 8: Dice and clDice per disease (3 seeds).

	Dice		clDice	
Disease	Base	Ours	Base	Ours
AMD	0.8643	**0.8671**	0.8884	**0.8978**
DR	0.8562	**0.8585**	0.8807	**0.8895**
Glaucoma	**0.7958**	0.7947	0.8178	**0.8227**
Normal	0.8363	**0.8541**	0.8635	**0.8904**

Thin vessels: thin-vessel recall 0.781 → 0.788 (139/200 images); Dice against the 1024 ground truth **0.857** → **0.880** (+2.2 points).

What worked

- Gains concentrate on thin vessels and connectivity – exactly the target.
- Connectivity improves in every disease group; Normal gains most.

What did not work (honest)

- **Flip TTA** lowered Dice for most models: they segment flipped images up to ∼2% worse, so averaging hurts. Validation now decides when to use it.
- **Glaucoma** is not improved – low contrast, fewer visible vessels.
- Two very poor-quality test images fail for every model (Dice < 0.1).
- Precision drops 0.4 points – the price of finding more vessels.
- The 1024 model needs ∼1.6× training time.

Conclusion, future work and contributions

Conclusion

- Replicated CAR-UNet on FIVES under a fair protocol; it beats U-Net, as the paper claims.
- Four targeted changes give **Dice 0.8436**, IoU 0.7379, clDice 0.8751 – better than the baseline on 9 of 11 metrics, all significant.
- Largest gains on thin vessels (+2.2 Dice points at 1024) and connectivity (+1.25).

Future work

- Focus on Glaucoma and low-quality images (FIVES quality labels).
- Full 2048×2048 training on a larger GPU.
- A lighter model (16 filters, as in the paper) for faster screening.

Table 9: Member contributions.

Member	Contribution
Abhishek Karthik J	**[e.g. model implementation, training]**
[Member 2]	**[e.g. data pipeline, preprocessing]**
[Member 3]	**[e.g. evaluation, ablation, figures]**
[Member 4]	**[e.g. report, literature review]**

References

[1] K. Jin *et al.*, "FIVES: A fundus image dataset for artificial intelligence based vessel segmentation," *Scientific Data*, 2022.

[2] C. Guo, M. Szemenyei, Y. Hu, W. Wang, W. Zhou, and Y. Yi, "Channel attention residual U-Net for retinal vessel segmentation," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021. arXiv:2004.03702.

[3] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 234–241, 2015.

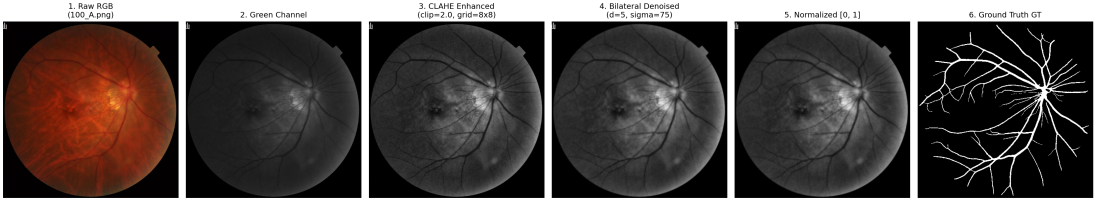
[4] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient channel attention for deep convolutional neural networks," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.

[5] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *European Conference on Computer Vision (ECCV)*, 2018.

[6] S. Shit, J. C. Paetzold, *et al.*, "clDice – a novel topology-preserving loss function for tubular structure segmentation," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

Thank you – questions?

Backup: preprocessing pipeline



- **Green channel:** haemoglobin absorbs green light most, so vessels have the highest contrast there.
- **CLAHE** (clip 2.0, 8×8 tiles): local contrast boost for unevenly lit images, clipped so noise is not amplified.
- **Bilateral filter** (d = 5, $\sigma = 75$): removes noise but keeps vessel edges sharp (a Gaussian blur would erase thin vessels).
- **FOV mask:** metrics are computed only inside the retinal circle.

Backup: complete metric comparison

Table 10: All metrics, baseline vs. modified (200 test images, 3 seeds; thin-vessel recall from seed 0).

Metric	Baseline	Modified	Change (pts)	Better on	Winner
Accuracy	0.9728	0.9737	+0.09	130/200	modified
Sensitivity	0.8310	0.8465	+1.55	159/200	modified
Specificity	0.9855	0.9846	−0.09	51/200	baseline
Precision	0.8557	0.8516	−0.40	68/200	baseline
Dice	0.8382	0.8436	+0.54	140/200	modified
IoU	0.7295	0.7379	+0.83	140/200	modified
AUC-ROC	0.9860	0.9878	+0.17	168/200	modified
AUC-PR	0.9202	0.9260	+0.58	163/200	modified
clDice	0.8626	0.8751	+1.25	175/200	modified
Thin-vessel recall	0.7806	0.7883	+0.77	139/200	modified
Dice vs. 1024 ground truth	0.8572	0.8796	+2.24	–	modified